

Notes: Baron and Xiong (QJE, 2017)

Credit Expansion and Neglected Crash Risk

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1 Big picture

- **Question.** Do bank equity investors recognize the crash risk created by credit booms, or do they remain too optimistic during periods of rapid credit expansion?
- **Setting.** The paper studies 20 developed economies over 1920–2012 and asks whether bank credit expansion predicts subsequent bank equity crashes and bank equity excess returns.
- **Core challenge.** It is already known that credit booms forecast banking crises and recessions. But that does *not* automatically imply that bank shareholders are irrational. If shareholders see the extra crash risk coming, they should mark down bank prices today and then earn a *higher* expected return going forward as compensation.
- **Main empirical strategy.** The paper combines:
 - probit regressions for future bank equity crash risk,
 - panel return-predictability regressions for mean excess returns,
 - threshold regressions for *large* credit booms and busts,
 - interactions with dividend yield, and
 - quantile regressions as an alternative way to measure tail risk.
- **Main result.** Bank credit expansion predicts a *higher probability of bank equity crashes* in the next one to three years, but despite this higher crash risk it also predicts *lower*, not higher, subsequent bank equity returns.
- **Magnitude.** A one-standard-deviation increase in credit expansion predicts about a **11.4 percentage point decline** in three-year-ahead bank equity excess returns. Conditional on credit expansion exceeding a country’s **95th percentile**, the predicted bank equity excess return over the next three years is -37.3% .
- **Distinct sentiment channel.** Credit expansion is almost uncorrelated with bank dividend yield, yet the return-predictability of credit expansion is especially strong when dividend yield is low. The paper interprets this as credit-boom sentiment being especially related to *neglect of tail risk*, whereas dividend yield captures broader equity-market sentiment.
- **Economic takeaway.** During credit booms, bank shareholders do not seem to price bank equity as if they fully understand the elevated crash risk. The evidence is most consistent with overoptimism or neglect of crash risk.

2 Data and measurement (key objects)

2.1 Country panel and sample construction

- The sample contains **20 developed economies** with quarterly observations from **1920 to 2012**.
- To be included, a country must be classified as an advanced economy and have at least **40 years** of data for both bank credit expansion and bank equity returns.
- The authors report that for **12 countries** the series are mostly complete from around 1920 onward, while for **8 countries** the data are mostly complete from around 1950 onward.

- The final consolidated quarterly sample has **4,155** observations.
- Because the paper forecasts one-, two-, and three-year-ahead returns, it deliberately uses **nonoverlapping horizons**. This leaves:
 - **957** observations for one-year-ahead regressions,
 - **480** for two-year-ahead regressions,
 - **316** for three-year-ahead regressions.

2.2 Credit expansion

The paper’s central predictor is the annualized three-year change in bank credit to GDP:

$$\Delta\left(\frac{\text{bank credit}}{\text{GDP}}\right)_t = \frac{\left(\frac{\text{bank credit}}{\text{GDP}}\right)_t - \left(\frac{\text{bank credit}}{\text{GDP}}\right)_{t-3}}{3}.$$

- **Bank credit** is credit from the banking sector to domestic households and nonfinancial corporations.
- The authors focus on *bank* credit, not total credit, because bank-balance-sheet expansion is the most direct risk to the banking sector itself and because long historical nonbank credit data are limited.
- The series are built by merging:
 - BIS “long series on credit to private non-financial sectors,” and
 - Schularick–Taylor historical bank-loan data.
- The variable is standardized within country, but in the predictive regressions this is done using **only past information** at each point in time, so there is no look-ahead bias.
- The paper prefers the *change* in credit/GDP rather than the *level* because changes capture the cyclical component and net new lending, whereas the level contains long-run structural trends.

2.3 Equity returns and crash indicators

- The main dependent variable is the **bank equity index** return for each country.
- Returns are defined as **log excess total returns**: price return plus dividend yield minus the three-month short rate.
- Bank equity index data come primarily from **Global Financial Data**, supplemented by Moody’s Banking Manuals, Compustat, Datastream, and other historical sources.
- The authors use broad market-capitalization-weighted bank indexes for each country.
- They construct one-, two-, and three-year-ahead excess returns by summing future quarterly log excess returns.
- A **crash indicator** equals 1 if the bank equity index has any quarterly excess return below -30% within the next one, two, or three years.

- A corresponding **boom indicator** equals 1 if the bank equity index has any quarterly excess return above +30% within the same horizon.
- In the raw quarterly data, bank equity returns below -30% occur in about **3.2%** of quarters and returns above +30% occur in about **1.1%** of quarters.

2.4 Controls and descriptive facts

The main control variables are:

- bank dividend yield,
- book-to-market,
- inflation,
- term spread,
- investment-to-capital.

Important descriptive facts from Tables I and II:

- Mean annualized bank equity excess return is **5.9%**; mean nonfinancial equity excess return is **6.4%**.
- Bank dividend yield averages **3.7%**.
- Mean credit expansion is **1.3 percentage points of GDP per year**.
- The **95th percentile** of credit expansion is **6.4 percentage points of GDP per year**; the **5th percentile** is **-3.2 percentage points**.
- Credit expansion is strongly correlated with other aggregate credit measures, but its correlation with bank dividend yield is near zero.
- This near-zero correlation is important because it suggests that credit-boom sentiment is not just the same thing as the usual low-dividend-yield equity sentiment.

3 Models and empirical specifications (all equations + interpretation)

3.1 Why returns are the key test

The paper's logic is simple but powerful.

- If credit expansion predicts future crashes, that *alone* does not imply shareholder irrationality.
- Rational shareholders could fully anticipate the crash risk, mark down bank prices today, and therefore require a *higher expected return* going forward.
- So the joint test is:
 - does credit expansion predict higher crash risk?

- and, conditional on that, does it predict higher or lower future mean returns?
- The combination of *higher crash risk plus lower expected return* is the central evidence for overoptimism or neglected crash risk.

3.2 Probit specification for future crash risk

The baseline crash-risk regression is

$$\Pr[Y_{i,t} = 1 \mid \text{predictors}_{i,t}] = \Phi(\alpha_i^K + \beta^K \text{predictors}_{i,t}),$$

where:

- $Y_{i,t} = 1$ if a crash occurs within the next K years, for $K \in \{1, 2, 3\}$,
- $\Phi(\cdot)$ is the standard normal cdf,
- predictors include credit expansion and, depending on the specification, the control variables.

Interpretation. The probit measures whether a rise in bank credit expansion shifts probability mass toward future bank equity crashes.

The paper also estimates the same specification with a boom indicator and looks at

$$\Pr(\text{crash}) - \Pr(\text{boom})$$

to check whether credit expansion mainly raises *negative skewness* rather than simply overall volatility.

3.3 Mean excess return regression

To study expected returns, the paper estimates

$$r_{i,t+K} - r_{i,t+K}^f = \alpha_i^K + \beta^K \text{predictors}_{i,t} + \varepsilon_{i,t},$$

again for $K \in \{1, 2, 3\}$.

- The dependent variable is a nonoverlapping future log excess return.
- Country fixed effects absorb average cross-country differences.
- The coefficient on credit expansion tells us whether a credit boom predicts higher or lower future average bank equity returns.

Interpretation. This is the main test of whether shareholders demanded a compensating premium for the extra crash risk.

3.4 Threshold regressions for large booms and contractions

To measure magnitudes in extreme episodes, the paper estimates

$$r_{i,t+K} - r_{i,t+K}^f = \alpha^K + \beta_x^K \mathbf{1}\{\text{credit expansion} > x\} + \varepsilon_{i,t},$$

where x is a percentile threshold such as the 90th, 95th, or 98th percentile.

Similarly, for large contractions:

$$r_{i,t+K} - r_{i,t+K}^f = \alpha^K + \beta_y^K \mathbf{1}\{\text{credit expansion} < y\} + \varepsilon_{i,t},$$

for low percentile thresholds y .

Interpretation. These regressions ask: how bad are future bank equity returns after an unusually large credit boom? This is where the economically striking -37.3% three-year figure comes from.

3.5 Interaction with dividend yield

To compare credit-boom sentiment with equity-market sentiment, the paper estimates

$$r_{i,t+K} - r_{i,t+K}^f = \alpha_i^K + \beta_1^K (\text{credit expansion})_{i,t} + \beta_2^K (\text{bank dividend yield})_{i,t} + \beta_3^K (\text{interaction})_{i,t} + \varepsilon_{i,t}.$$

The interaction is modeled in two ways:

- a standard linear interaction,
- a nonlinear interaction that lets the effect vary by dividend-yield quintile.

Interpretation. If low dividend yield captures broad optimism and credit expansion captures tail-risk neglect, then the two should reinforce one another. The paper finds exactly that.

3.6 Quantile regressions as an alternative crash-risk measure

The paper also estimates quantile regressions:

$$\text{Quantile}_q(r_{i,t+K} - r_{i,t+K}^f \mid \text{predictors}_{i,t}) = \alpha_{i,q}^K + \beta_q^K \text{predictors}_{i,t}.$$

Two alternative crash-risk objects are built from these regressions.

First,

$$\beta^{\text{difference}} = \beta^{\text{median}} - \beta^{\text{mean}},$$

which measures how much mean returns are pulled down by tail events relative to median returns.

Second, the negative-skewness statistic:

$$\beta^{\text{negative skew}} = (\beta^{q=50} - \beta^{q=5}) - (\beta^{q=95} - \beta^{q=50}).$$

Interpretation.

- If the left tail worsens more than the right tail improves, future returns become more negatively skewed.
- This provides a threshold-free robustness check for the probit crash-risk results.

3.7 Econometric design choices

The paper makes several deliberate design choices:

- **Nonoverlapping returns.** This avoids the standard overlapping-returns inference problems.
- **Dual clustering by country and time.** This allows for both global shocks and persistent country-specific shocks.
- **Past-information-only normalization.** Predictor variables are normalized using only data available at the time.
- **Historical-episode aggregation.** As a robustness check, large credit booms are collapsed into distinct global or regional episodes before testing average returns.

4 Identification and interpretation (what makes the estimates credible)

4.1 What the paper needs to separate

The paper needs to distinguish four different things:

- credit expansion as a predictor of future *crashes*,
- credit expansion as a predictor of future *mean returns*,
- ordinary time-varying equity premia captured by standard controls,
- broad equity-market optimism captured by dividend yield.

Only after separating these can the paper say something about investor beliefs.

4.2 Why lower future returns are evidence of overoptimism

The key identifying logic is an asset-pricing argument:

- If shareholders are rational and perceive more crash risk, they should demand a higher expected return.
- That higher expected return would show up as lower prices *today* and higher average returns *tomorrow*.
- Instead, the paper finds that higher credit expansion predicts both more crash risk and lower future returns.

Interpretation. This joint pattern is hard to reconcile with fully rational pricing of bank crash risk.

4.3 Why credit expansion is distinct from dividend-yield sentiment

The paper shows that:

- credit expansion and bank dividend yield are only weakly correlated,
- dividend yield does not strongly predict bank crash risk,
- credit expansion does predict crash risk,
- but the return-predictability of credit expansion is stronger when dividend yield is low.

Interpretation. Credit expansion appears to capture something different from ordinary equity-market sentiment. The paper interprets this as a neglected-tail-risk channel.

4.4 Why the estimates are not driven by one episode

The authors address the concern that the results might just be the Great Depression or the 2007–2008 crisis.

- They rerun the analysis on geographical subsamples.
- They rerun it on the **1950–2003** subsample.
- They cluster large credit booms into **19 historical episodes**.

The main negative-return and crash-risk results survive these exercises.

4.5 Limits of interpretation

- The paper does not structurally estimate investor beliefs. It infers them from the joint pattern of future crash risk and future returns.
- The authors are careful to note that some of the lower future returns could reflect elevated risk appetite, but *sharply negative* predicted excess returns during large booms are much harder to rationalize that way.
- The focus is on **bank equity**, because the banking sector directly bears the balance-sheet risk from credit expansion.

5 Tables and figures

5.1 Figure I and Tables I–II: what the data look like

Read:

- Figure I shows that credit expansion is highly cyclical and mean-reverting across the 20 countries.
- Table I gives the descriptive scale of the problem:
 - mean bank equity excess return is **5.9%**,
 - bank dividend yield averages **3.7%**,
 - mean credit expansion is **1.3 percentage points of GDP**,
 - the **95th percentile** of credit expansion is **6.4 points of GDP**.

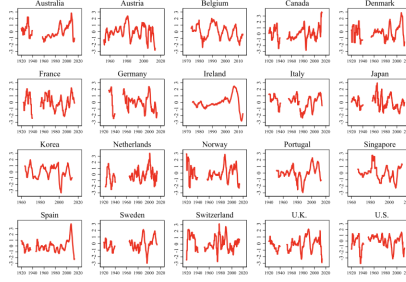


FIGURE I

Credit Expansion

Credit expansion, measured as the past three-year change of bank credit to GDP and denoted $\Delta(\frac{\text{bank credit}}{\text{GDP}})$, is plotted over time for the 20 countries in the sample. Observations are quarterly over the period 1920–2012 and in units of standard deviations. Bank credit refers to credit issued by banks to domestic households and domestic private nonfinancial corporations.

- Table II is important because it shows that credit expansion is strongly related to other credit aggregates but is almost unrelated to dividend yield.
- This near-zero relation to dividend yield is one of the key pieces of evidence that credit expansion is not simply a relabeling of standard equity-market sentiment.

TABLE II
CORRELATIONS

Correlation of $\Delta(\frac{\text{bank credit}}{\text{GDP}})$ and:	Average correlation	Std. err.
$\Delta(\frac{\text{total credit}}{\text{GDP}})$	.792***	(.048)
$\Delta(\frac{\text{total credit to HHs}}{\text{GDP}})$	.636***	(.054)
$\Delta(\frac{\text{total credit to private NFCs}}{\text{GDP}})$	.608***	(.067)
$\Delta(\frac{\text{bank assets}}{\text{GDP}})$	.592***	(.056)
Growth of household housing assets	.316***	(.085)
$\Delta(\frac{\text{gross external liabilities}}{\text{GDP}})$	.338***	(.073)
Current account deficit GDP	.172***	(.057)
Market dividend yield	-.026	(.046)
Bank dividend yield	.052	(.046)
Book to market	-.094*	(.056)
Inflation	-.103***	(.039)
Term spread	-.136***	(.049)
Investment to Capital	.300***	(.070)

Notes. This table reports correlations of the past three-year change in bank credit to GDP with various other measures of aggregate credit and with the control variables (market dividend yield, year-over-year inflation, term spread, book to market, and nonresidential investment to capital). Because the measurement of these variables may be different from country to country, each correlation is first calculated country by country; then, the correlation coefficient is averaged (and standard errors are calculated) across the 20 countries. *, **, and *** denote statistical significance at 10%, 5%, and 1% levels, respectively. Observations are quarterly over the sample of 20 countries, 1920–2012.

	N	Mean	Median	Std. dev.	1%	5%	10%	90%	95%	99%	Average cross-country correlation (with U.S.)
Quarterly log returns, annualized											
Bank index: excess total returns	4,155	0.059	0.045	0.286	-1.376	-0.762	-0.507	0.587	0.857	1.803	0.394
Bank index: dividend yield	4,155	0.037	0.036	0.019	0.000	0.008	0.014	0.060	0.067	0.093	0.305
Nonfinancials index: excess total returns	4,092	0.064	0.060	0.256	-1.266	-0.748	-0.518	0.627	0.856	1.461	0.411
Market index: dividend yield	4,092	0.036	0.033	0.020	0.008	0.013	0.016	0.059	0.068	0.117	0.606
Credit to private households and nonfinancial corporations, past three-year annualized percentage-point change											
$\Delta(\frac{\text{bank credit}}{\text{GDP}})$	4,155	0.013	0.011	0.032	-0.059	-0.032	-0.022	0.050	0.064	0.115	0.221
Control variables											
Inflation	4,147	0.037	0.028	0.043	-0.076	-0.011	0.001	0.090	0.119	0.185	0.686
Term spread	4,088	0.012	0.012	0.018	-0.042	-0.016	-0.007	0.080	0.036	0.053	0.184
Book to market	2,437	0.707	0.621	0.416	0.265	0.341	0.377	1.042	1.333	2.564	0.543
Investment to Capital	3,266	0.102	0.099	0.019	0.068	0.075	0.081	0.127	0.140	0.161	0.550

Notes. Summary statistics are reported for log total excess returns for both the bank and nonfinancials equity indexes. Summary statistics are also reported for the past three-year change in bank credit to GDP and the control variables. All statistics are pooled across countries and time. Observations are quarterly over the sample of 20 countries, 1920–2012.

5.2 Figure II and Table III: crash risk rises after credit booms

Read:

- Figure II is the clean visual summary of the paper:
 - bank equity rises strongly before the peak of a credit boom,
 - then falls sharply afterward,
 - with the decline continuing for roughly three years.
- The average peak-to-trough decline in the bank equity index is over **30%** in log terms.
- Table III formalizes this with probit marginal effects:
 - a one-standard-deviation rise in credit expansion raises the probability of a future bank equity crash by about **2.7** percentage points at the one-year horizon,

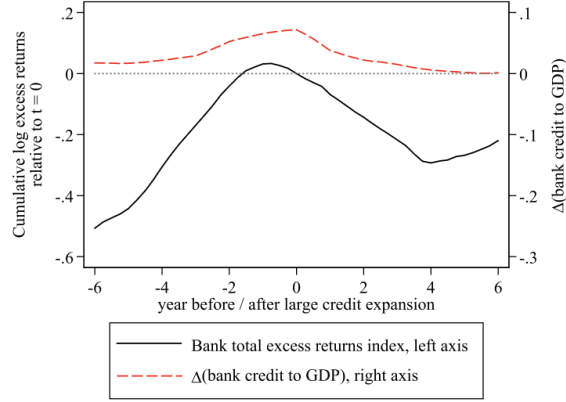


FIGURE II

Bank Equity Prices and Bank Credit before and after Large Credit Expansions

The past three-year change in bank credit to GDP (i.e., $\Delta(\frac{\text{bank credit}}{\text{GDP}})$) and the bank total excess log returns index are plotted before and after a large credit expansion. A large credit expansion is defined as *credit expansion* exceeding the 95th percentile threshold, which is calculated for each country and each point in time using only past information to avoid any future-looking bias. $\Delta(\frac{\text{bank credit}}{\text{GDP}})$ and bank total excess log returns are pooled averages across time and countries, conditional on the given number of years before or after the start of a banking crisis. The average bank log returns are then cumulated from $t = -6$ to $t = +6$, and the level is adjusted to be 0 at $t = 0$. Observations are over the sample of 20 countries, 1920–2012.

- **3.3** points at the two-year horizon,
- **5.4** points at the three-year horizon.

- These effects remain after adding dividend yield and the five standard controls.
- The crash-minus-boom comparison is even more revealing: credit expansion raises the probability of *negative* tail events more than positive ones, so the result is about negative skewness, not just higher volatility.

5.3 Table IV and Table V: mean returns become lower, not higher

	1 year ahead			2 years ahead			3 years ahead		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
$\Delta(\frac{\text{bank credit}}{\text{GDP}})$	-0.032**	-0.034**	-0.035**	-0.060**	-0.061**	-0.057**	-0.114**	-0.113**	-0.106**
	[-2.146]	[-2.295]	[-2.365]	[-3.455]	[-3.355]	[-3.150]	[-3.655]	[-3.606]	[-3.296]
log(bank div. yield)	0.040**	0.042**	0.042*	0.069**	0.070**	0.067**	0.111**	0.117**	0.115**
	[2.158]	[2.257]	[1.840]	[2.468]	[2.568]	[2.236]	[3.818]	[4.682]	[3.842]
Inflation		-0.184	[-0.970]		-0.011	[-0.040]		0.015	[0.042]
Term spread		0.019	[0.718]		0.024	[0.742]		0.099	[1.763]
log(book to market)		0.030	[0.718]		0.046	[0.742]		0.083	[1.763]
log(investment to capital)		0.015	[0.718]		0.002	[0.742]		0.016	[0.307]
R^2	0.028	0.028	0.048	0.057	0.064	0.06	0.097	0.131	0.102
Adj. R^2	0.007	0.008	0.026	0.031	0.023	0.019	0.055	0.072	0.041
N	957	957	957	957	480	480	480	316	316

Notes. This table reports estimates from the panel regression with fixed effects model specified in equation (2) for the bank equity index. The dependent variable is log excess total returns, which is regressed on $\Delta(\frac{\text{bank credit}}{\text{GDP}})$ and several subsets of control variables known to predict the equity premium. Explanatory variables are in standard deviation units. Returns are nonoverlapping at one-, two-, and three-year-ahead horizons. A coefficient of -0.032 means that a one-standard deviation increase in $\Delta(\frac{\text{bank credit}}{\text{GDP}})$ predicts a 3.2% decrease in subsequent returns. Analogous results for the nonfinancials equity index are reported in Online Appendix Table III. t-statistics in brackets are computed from standard errors daily clustered on country and time. *, **, and *** denote statistical significance at 10%, 5%, and 1% levels, respectively.

	1 year ahead			2 years ahead			3 years ahead		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Crash dummy	Boom dummy	Difference	Crash dummy	Boom dummy	Difference	Crash dummy	Boom dummy	Difference
No controls									
$\Delta(\frac{\text{bank credit}}{\text{GDP}})$	0.027**	-0.003	0.030**	0.033**	-0.002	0.035**	0.054**	-0.012**	0.066**
	[2.40]	[-0.446]	[2.24]	[3.11]	[-0.27]	[3.04]	[4.27]	[-2.91]	[5.20]
N	957	957	957	480	480	480	316	316	316
No controls									
log(bank div. yield)	-0.015	0.005	-0.020	-0.022	0.009	-0.030	-0.020	0.005	-0.024
	[-1.16]	[0.86]	[-1.28]	[-1.30]	[1.44]	[-1.37]	[-0.98]	[0.46]	[-0.87]
N	957	957	957	480	480	480	316	316	316
With bank div. yield as control									
$\Delta(\frac{\text{bank credit}}{\text{GDP}})$	0.029**	-0.003	0.032**	0.034**	-0.002	0.036**	0.054**	-0.012**	0.066**
	[2.54]	[-0.59]	[2.49]	[3.04]	[-0.30]	[2.95]	[4.17]	[-3.02]	[5.13]
log(bank div. yield)	-0.015	0.006	-0.023	-0.023	0.009	-0.032	-0.021	0.005	-0.026
	[-1.33]	[1.01]	[-1.47]	[-1.39]	[1.48]	[-1.46]	[-1.09]	[0.49]	[-0.96]
N	957	957	957	480	480	480	316	316	316
With all five controls									
$\Delta(\frac{\text{bank credit}}{\text{GDP}})$	0.026**	-0.003	0.029**	0.027**	-0.002	0.028*	0.046**	-0.013**	0.059**
(coeff. on controls not reported)	[3.03]	[-0.66]	[2.96]	[2.21]	[-0.29]	[1.80]	[3.11]	[-3.24]	[3.48]
N	957	957	957	480	480	480	316	316	316

Notes. This table reports estimates from the probit regression model specified in equation (3) for the bank equity index in subsequent one, two, and three years. The main dependent variable is the crash indicator ($I = I_{\text{crash}}$), which takes on a value of 1 if there is a future equity crash, defined as a quarterly drop of -30% or more, in the next K years ($K = 1, 2$, and 3) and 0 otherwise. The crash indicator is regressed on $\Delta(\frac{\text{bank credit}}{\text{GDP}})$ and several subsets of control variables known to predict the equity premium. Explanatory variables are in standard deviation units. All reported estimates are marginal effects. A coefficient of 0.027, for example, means that a 1 standard deviation increase in $\Delta(\frac{\text{bank credit}}{\text{GDP}})$ predicts a 2.7% increase in the likelihood of a future crash. This table also reports estimates from equation (3) with $I = \text{boom}$, a symmetrically defined right-tail event, along with the difference in the marginal effects between the two probit regressions (the probability of a crash minus the probability of a boom). Analogous results for the nonfinancials equity index are reported in Online Appendix Table III. t-statistics in brackets are computed from standard errors daily clustered on country and time. *, **, and *** denote statistical significance at 10%, 5%, and 1% levels, respectively. Observations are over the sample of 20 countries, 1920–2012.

	Bank equity index returns subsequent to $\Delta(\frac{\text{bank credit}}{\text{GDP}})$ being:									
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	<-2%	<-5%	<-10%	<-25%	<-50%	>50%	>75%	>90%	>95%	>98%
One-year-ahead returns $E[r - r_f]$	0.074	0.126**	0.077**	0.059**	0.049**	-0.016	-0.042	-0.073	-0.094	-0.061
	[1.185]	[2.216]	[1.987]	[2.256]	[2.058]	[-0.385]	[-0.767]	[-0.812]	[-0.918]	[-1.292]
Adj. R^2	0.002	0.009	0.005	0.006	0.01	0.01	0.012	0.011	0.01	0.004
# obs. meeting threshold	51	72	110	235	464	493	271	121	80	44
Two-year-ahead returns $E[r - r_f]$	0.146*	0.19**	0.164**	0.128**	0.092**	-0.021	-0.077	-0.155*	-0.179**	-0.133*
	[1.697]	[2.575]	[3.958]	[3.018]	[2.52]	[-0.325]	[-0.904]	[-1.729]	[-2.021]	[-1.951]
Adj. R^2	0.004	0.011	0.012	0.016	0.017	0.017	0.027	0.028	0.022	0.008
# obs. meeting threshold	34	35	54	118	227	253	139	60	40	23
Three-year-ahead returns $E[r - r_f]$	0.232**	0.283**	0.264**	0.208**	0.179**	-0.075	-0.125	-0.24**	-0.373**	-0.561**
	[2.298]	[3.644]	[2.846]	[4.406]	[3.022]	[-0.841]	[-1.215]	[-2.384]	[-3.522]	[-5.857]
Adj. R^2	0.008	0.018	0.023	0.03	0.039	0.009	0.047	0.04	0.041	0.048
# obs. meeting threshold	18	25	36	73	147	169	99	38	19	11

Notes. This table reports average log excess returns of the bank equity index subsequent to large credit expansions (when $\Delta(\frac{\text{bank credit}}{\text{GDP}})$ exceeds a given percentile threshold) and subsequent to large credit contractions (when $\Delta(\frac{\text{bank credit}}{\text{GDP}})$ falls below a given percentile threshold). Estimates, along with corresponding t-statistics and adjusted R^2 values, are computed using regression models (3) and (4) with nonoverlapping one-, two-, and three-year-ahead returns. To avoid any future-looking bias, percentile thresholds are calculated for each country and each point in time using only past information. t-statistics in brackets are computed from standard errors daily clustered on country and time. Analogous results for the nonfinancials equity index are reported in Online Appendix Table III. *, **, and *** denote statistical significance at 10%, 5%, and 1% levels, respectively. Observations are over the sample of 20 countries, 1920–2012.

Read:

- Table IV is the central pricing result.
- A one-standard-deviation increase in credit expansion predicts:
 - -3.2% lower one-year-ahead excess returns,
 - -6.0% lower two-year-ahead excess returns,
 - -11.4% lower three-year-ahead excess returns.

- This is the opposite sign from what rational crash-risk pricing would predict.
- Table V shows the nonparametric magnitude for *large* booms:
 - if credit expansion exceeds the **95th percentile**, the predicted bank equity excess return is about
 - * -9.4% over one year,
 - * -17.9% over two years,
 - * -37.3% over three years.
- The two- and three-year numbers are statistically significant and economically huge.
- Table V also shows the asymmetry with contractions: large credit contractions are followed by positive bank equity returns. So the bad-news result is concentrated in the boom state, exactly where the overoptimism story says it should be.

5.4 Figure III and Table VI: credit expansion is most powerful when dividend yield is low

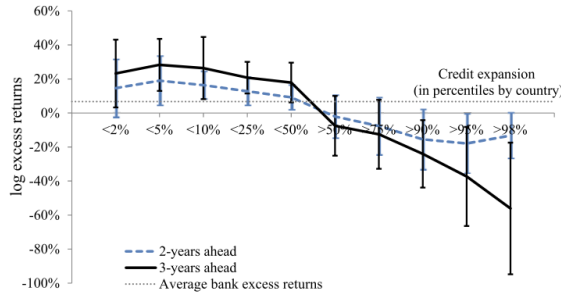


FIGURE III
Bank Equity Index Returns Subsequent to Large Credit Expansions and Contractions

This figure plots estimates reported in Table V. The plot shows the magnitude of bank equity index excess returns two and three years subsequent to large credit expansions (defined as when $\Delta(\frac{\text{bank credit}}{\text{GDP}})$ exceeds a given percentile threshold), in addition to average returns subsequent to large credit contractions (when $\Delta(\frac{\text{bank credit}}{\text{GDP}})$ falls below a given percentile threshold). To avoid any future-looking bias, percentile thresholds are calculated for each country and each point in time using only past information. Average returns conditional on the thresholds are computed using regression models (3) and (4) with nonoverlapping returns. 95% confidence intervals are computed using dually clustered standard errors. Observations are over the sample of 20 countries, 1920–2012.

	1 year ahead				2 years ahead				3 years ahead			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
$\Delta(\frac{\text{bank credit}}{\text{GDP}})$	-0.032**		-0.034**	-0.035***	-0.060***		-0.061***	-0.057***	-0.114***		-0.119***	-0.106***
	[-2.146]		[-2.285]	[-2.985]	[-3.455]		[-3.355]	[-3.150]	[-3.655]		[-3.609]	[-3.226]
log(bank div. yield)	0.040**	0.042**	0.042*			0.069**	0.070**	0.067**		0.111***	0.117***	0.115***
	[2.158]	[2.257]	[1.840]			[2.468]	[2.568]	[2.236]		[3.818]	[4.682]	[3.842]
Inflation			-0.184				-0.011					0.015
			[-0.970]				[-0.040]					[0.042]
Term spread			0.019				0.024					0.099*
			[0.718]				[0.742]					[1.783]
log(book to market)			0.030				0.046					0.081
			[0.792]				[0.782]					[1.037]
log(investment to capital)			0.015				0.002					0.016
			[0.641]				[0.075]					[0.307]
R^2	0.028	0.028	0.048	0.057	0.064	0.06	0.097	0.104	0.131	0.102	0.194	0.233
Adj. R^2	0.007	0.008	0.026	0.031	0.023	0.019	0.055	0.055	0.072	0.041	0.137	0.167
N	957	957	957	957	480	480	480	480	316	316	316	316

Notes: This table reports estimates from the panel regression with fixed effects model specified in equation (2) for the bank equity index. The dependent variable is log excess total returns, which is regressed on $\Delta(\frac{\text{bank credit}}{\text{GDP}})$ and several subsets of control variables known to predict the equity premium. Explanatory variables are in standard deviation units. Returns are nonoverlapping at one-, two-, and three-year-ahead horizons. A coefficient of -0.032 means that a one standard deviation increase in $\Delta(\frac{\text{bank credit}}{\text{GDP}})$ predicts a 3.2% decrease in subsequent returns. Analogous results for the nonfinancial equity index are reported in Online Appendix Table III. *statistic in brackets are computed from standard errors dually clustered on country and time. *, **, and *** denote statistical significance at 10%, 5%, and 1% levels, respectively.

Read:

- Figure III plots predicted future bank equity returns after large booms and contractions over different percentile thresholds. The return profile becomes sharply negative in the right tail of credit expansion.
- Table VI shows that the interaction between credit expansion and dividend yield is nonlinear.
- The predictive power of credit expansion is weak when dividend yield is high, but much stronger when dividend yield is in its **lowest quintile**.
- In the nonlinear interaction specification, a one-standard-deviation increase in credit expansion predicts an additional -3.9% one-year-ahead return and -14.4% three-year-ahead return when dividend yield is in the lowest quintile relative to the highest quintile.

- This is exactly the paper’s interpretation: credit-boom sentiment and low-dividend-yield sentiment are distinct, but they reinforce one another.

5.5 Table VII and Figure IV: the result survives historical-episode clustering

TABLE VII
BANK EQUITY INDEX RETURNS SUBSEQUENT TO LARGE EXPANSIONS: GROUPED BY
HISTORICAL EPISODES

				Returns on bank equity			
				(1)	(2)	(3)	
Episode	Associated crisis	Year: quarter	Country	1 year ahead	2 years ahead	3 years ahead	
1	Great Depression	1929:1	France	−0.119	−0.338	−0.632	
		1932:4	United States	−0.353	−0.173	0.244	
2		1958:4	Japan	0.105	0.211	0.135	
3		1960:4	United Kingdom	0.243	0.141	0.097	
4	Secondary banking crisis	1962:4	Japan	0.268	0.243	0.461	
5		1969:2	Sweden	−0.405	−0.177	−0.193	
6		1972:4	United Kingdom	−0.453	−1.457	−0.708	
7		1974:1	United States	−0.384	−0.147	−0.140	
8		1977:4	Switzerland	−0.044	0.105	0.158	
9		1979:2	Belgium	−0.271	−0.656	−0.498	
		1980:4	Netherlands	−0.211	−0.250	−0.024	
10		1981:1	Ireland	−0.429	−0.245	0.269	
		1981:3	Canada	−0.181	0.237	0.057	
		1982:4	United Kingdom	0.305	0.453	0.587	
11	Savings and Loan crisis	1986:4	United States	−0.273	−0.108	0.012	
12	Scandinavian financial crises	1986:3	Denmark	0.004	−0.116	−0.141	
		1986:4	Sweden	−0.170	0.197	0.215	
		1987:4	Norway	−0.253	−0.062	−0.734	
13	Japanese financial crisis	1987:2	Japan	−0.105	−0.062	−0.206	
14		1987:3	Australia	0.108	0.034	−0.287	
15		1989:1	Belgium	−0.124	−0.231	−0.211	
16		1994:3	Korea	−0.162	−0.502	−1.096	
17		1997:1	Netherlands	0.408	0.304	0.464	
		1997:2	Ireland	0.661	0.533	0.293	
		1998:3	Portugal	0.074	0.282	−0.026	
		1999:2	Spain	0.096	0.071	−0.143	
18		East Asian crisis	1997:4	Korea	−0.119	−0.225	−0.923
19		Great Recession	2004:1	Spain	0.130	0.415	0.542
	2004:3		Ireland	0.263	0.430	0.279	
	2005:2		Denmark	0.234	0.330	−0.156	
	2006:3		Australia	0.136	−0.243	−0.006	
	2006:4		United States	−0.253	−0.727	−0.701	
	2007:2		Canada	−0.234	−0.184	−0.045	
	2007:3		France	−0.401	−0.476	−0.574	
	2007:3		Sweden	−0.465	−0.392	−0.254	
		2007:4	Italy	−0.813	−0.566	−0.896	
		2008:4	Portugal	0.164	−0.165	−1.123	
Average bank equity index returns over episodes				−0.099	−0.136	−0.180	
				[−1.945]	[−1.524]	[−1.993]	
N (episodes)				19	19	19	

Notes. This table presents an alternative method of calculating average bank equity returns subsequent to large credit expansions, along with standard errors, taking into account correlations across countries and over time. It lists one-, two-, and three-year-ahead returns of the bank equity index subsequent to the initial quarter of all large credit expansions, defined as $\Delta(\frac{\text{bank equity}}{\text{GDP}})$ exceeding a 95th percentile threshold within each country. To avoid any future-looking bias, percentile thresholds are calculated at each point in time using only past information. Then, concurrent observations of large credit expansions across countries are clustered into distinct historical episodes (e.g., the Great Depression, the East Asian crisis, the 2007–2008 global financial crisis). Returns from the resulting historical episodes are first averaged within each historical episode; then, an average and *t*-statistic are calculated across historical episodes, taking each distinct historical episode as a single, independent observation. Observations are over the sample of 20 countries, 1920–2012.

f4.png

Read:

- Table VII groups large credit booms into **19 distinct historical episodes**, including the Great Depression, the Scandinavian crises, the Japanese crisis, the East Asian crisis, and the Great Recession.
- After averaging within episodes and treating each episode as one independent observation, the average bank equity return is still negative:
 - about −9.9% over one year,
 - −13.6% over two years,
 - −18.0% over three years.
- Figure IV shows the cross-episode three-year-ahead returns visually. Many large credit booms are followed by very poor bank equity performance, not just one or two famous crises.

5.6 Tables VIII–X: subsample robustness and alternative tail measures

TABLE IX
FURTHER ROBUSTNESS IN GEOGRAPHICAL AND TIME SUBSAMPLES

		3-year ahead bank equity index excess returns equivalent to $\Delta \ln(\text{bank credit}/\text{GDP})$: between		
		(1)	(2)	(3)
		>-90%	>90%	>98%
Full sample	$Etr - E_{tr}$	-0.24**	-0.373**	-0.561**
	R^2	[2.284]	[2.522]	[2.857]
	N	0.04	0.04	0.04
United States	$Etr - E_{tr}$	-0.435	-1.701	
	R^2	[1.527]	[1.743]	
	N	0.126	0.146	
English-speaking countries	$Etr - E_{tr}$	-0.11	-0.166	-0.298**
	R^2	[-0.087]	[-0.73]	[-12.843]
	N	0.021	0.042	0.036
Western Europe	$Etr - E_{tr}$	-0.302**	-0.369**	-0.561**
	R^2	[-2.194]	[-2.314]	[-2.808]
	N	0.046	0.038	0.059
Southern Europe	$Etr - E_{tr}$	-0.235	-0.282**	-0.282**
	R^2	[-1.082]	[-1.172]	[-0.728]
	N	0.033	0.018	0.018
Scandinavia	$Etr - E_{tr}$	-0.553**	-0.6***	-0.733**
	R^2	[-2.647]	[-5.877]	[-14.362]
	N	0.068	0.055	0.071
1950-2003, all countries	$Etr - E_{tr}$	-0.187**	-0.174*	-0.292**
	R^2	[-2.345]	[-1.775]	[-4.198]
	N	0.042	0.022	0.027

TABLE X
QUANTILE REGRESSIONS AS AN ALTERNATIVE MEASURE OF CRASH RISK

		(1)	(2)	(3)
		1 year ahead	2 years ahead	3 years ahead
$\Delta(\frac{bush}{GDP})$	Mean	-0.632**	-0.0607**	-0.114**
		(-2.15)	(-3.66)	(-3.65)
	Median	-0.019**	-0.074**	-0.041**
		(-3.14)	(-3.08)	(-4.21)
Difference	ρ -value	0.014	0.200	0.037*
		0.014		0.024
$\Delta(\frac{bush}{GDP})$	q = 5	-0.075**	-0.143	-0.124
		(-3.81)	(-1.05)	(-1.59)
	q = 50	-0.019**	-0.034	-0.077**
		(-2.67)	(-3.08)	(-4.11)
q = 95		-0.028*	-0.067**	-0.121**
		(-2.19)	(-2.47)	(-2.15)
Negative skew		0.065**	0.018	0.083**
		(2.86)	(0.68)	(2.15)
<i>N</i>		967	480	160

[illegible]TABLE VIII
GEOGRAPHICAL AND TIME RANGES

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Full sample	U.S.	English-speaking	W. Europe	S. Europe	Scandinavia	1950-2010, all countries
Panel A: Profit estimation of the four-panel rank index regressions							
$\Delta \ln \text{Profit}^{(1)}$	0.066***	0.112	0.074	0.049**	0.102**	0.041	0.048**
$\Delta \ln \text{Profit}^{(2)}$	0.477***	0.477***	0.477***	0.477***	0.477***	0.477***	0.477***
N	338	24	87	231	82	87	238
$\Delta \ln \text{Profit}^{(3)}$	0.000	0.000	0.000	0.000	0.000	0.000	0.000
$\Delta \ln \text{Profit}^{(4)}$	0.311	0.000	0.004	0.313	0.640	0.010	0.170
$\Delta \ln \text{Profit}^{(5)}$	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Panel B: OLS estimation of the four-panel rank index regressions							
$\Delta \ln \text{Profit}^{(1)}$	-0.114***	-0.000	-0.050*	-0.153**	-0.193**	-0.194**	-0.114**
$\Delta \ln \text{Profit}^{(2)}$	-1.400***	-1.503***	-1.400***	-1.444***	-1.703***	-1.400***	-1.400***
$\Delta \ln \text{Profit}^{(3)}$	0.070	0.070	0.070	0.070	0.070	0.070	0.070
N	338	24	87	231	82	87	238
$\Delta \ln \text{Profit}^{(4)}$	-0.186***	0.000	-0.047	-0.176***	-0.270**	-0.252***	-0.091***
$\Delta \ln \text{Profit}^{(5)}$	0.167***	0.311***	0.172	0.133***	0.116	0.27	0.231

Notes: This table demonstrates that the estimates reported in Tables III and IV for the probit (Panel A) and OLS (Panel B) regression models are robust within various geographic and time subsets. Time subsets are 1930–1932 (the full sample) and 1950–2003 (i.e., excluding both the 2007–2008 financial crisis and the Great Depression). The table reports estimates—using the same methodology as in Tables III and IV—of future log excess returns of the bank equity index. Within Panels A and B, the probit and OLS coefficients are estimated with (yes) or without (no) the five standard controls. Coefficients reported in this table are on $\Delta \ln \text{RET}_{i,t+1}$; coefficients on control variables are omitted. * , ** , and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Notes: This table demonstrates that the estimates reported in Tables V for the nonparametric regression model are robust within various geographical and time subsets. Time subsets are: 1920–2012 (the full sample) and 1950–2003 (i.e., excluding both the 2007–2008 financial crisis and the Great Depression). The table reports the three-year-ahead bank index returns subsequent to large credit expansions in various time and geographical subsets. *, **, and *** denote statistical significance at 10%, 5%, and 1% levels, respectively.

Read:

- Table VIII shows that both the crash-risk and lower-mean-return results remain in geographical subsamples and in the **1950–2003** subsample.
- Table IX shows that the negative-return result after large credit booms also remains in those subsamples, though precision falls because sample sizes get small.
- Table X gives the quantile-regression robustness check.
 - Median returns still fall after credit expansion:

$$-0.019, \quad -0.041, \quad -0.077$$
 at one-, two-, and three-year horizons.
 - Mean returns fall even more, so the difference between mean and median picks up additional crash risk.
 - The negative-skewness statistic is positive, especially at the one- and three-year horizons.
- The interpretation is that credit expansion does not just worsen average returns; it makes the left tail more dangerous.

6 Short summary

- **Method.** The paper uses a long international panel of bank credit and bank equity returns, together with probit, OLS, threshold, interaction, and quantile regressions.
- **Identification logic.** If shareholders rationally price the extra crash risk from credit booms, future expected returns should rise, not fall. Therefore, higher crash risk plus lower future returns is evidence of overoptimism or neglected tail risk.
- **Application.** In 20 developed economies from 1920 to 2012, bank credit expansion strongly predicts subsequent bank equity crashes.

- **Main result.** Credit expansion also predicts lower mean bank equity excess returns. After a very large boom (above the 95th percentile), the next three years of bank equity returns are predicted to be about -37.3% . Credit-boom sentiment is distinct from dividend-yield sentiment, but the two reinforce each other when dividend yield is low.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper establishes that credit booms forecast both **higher bank equity crash risk** and **lower bank equity excess returns**.
- That joint pattern is inconsistent with shareholders fully pricing the extra crash risk associated with credit expansions.
- The evidence therefore supports the Minsky–Kindleberger view that overoptimism is an important driver of credit booms.

7.2 Economic intuition

- During a credit boom, bank balance sheets expand and tail risk rises.
- If shareholders understood that fully, current bank prices should fall enough to make future returns high on average.
- Instead, future bank returns are low, especially after large booms. So prices at boom time appear too high relative to the subsequent crash risk.
- The fact that the effect is strongest when dividend yield is also low suggests that general optimism and tail-risk neglect interact with one another.

7.3 Takeaway

This paper is important because it converts a broad historical idea—that credit booms are fueled by overoptimism—into a sharp asset-pricing test. The striking result is not only that credit expansion predicts crises, but that bank shareholders *do not earn compensation* for bearing that extra crash risk. In the data, credit booms are periods when bank equity looks especially dangerous *and* especially overvalued.